Cover page

مركز الأبحاث المشترك للذكاء الاصطناعي

Joint Research Center for AI

SDAIA | KFUPM

King Fahd University of Petroleum and Minerals (KFUPM)

ASSIGNMENT COVER SHEET

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Table of Contents

1. Abstract

Artificial intelligence (AI) has become an important technology for improving intelligent traffic management by enabling more accurate traffic prediction, congestion management, and decision-making. This literature review examines recent research on AI-based intelligent traffic management systems to identify current trends, commonly used techniques, and existing research gaps. The selected studies were identified through a literature search conducted using Google Scholar with keywords related to intelligent traffic management, machine learning, deep learning, Internet of Things (IoT), traffic prediction, and smart transportation. Ten peer-reviewed papers published in the last 5 years were selected based on their relevance to the topic. The review found that machine learning, deep learning, Graph Neural Network (GNN), Long Short-Term Memory (LSTM), reinforcement learning, and Explainable Artificial Intelligence (XAI) are widely used to improve traffic forecasting, congestion prediction, accident detection, and traffic signal optimization. The studies also highlighted the growing role of IoT technologies in supporting real-time traffic monitoring and intelligent decision-making. Despite these advancements, common challenges remain, including the need for high-quality real-time data, computational complexity, infrastructure requirements, and limited real-world implementation. Overall, the reviewed literature demonstrates that AI has significant potential to improve transportation efficiency and support the development of sustainable smart cities. Future research should focus on developing more scalable, interpretable, and practical AI solutions that can be successfully implemented in real-world traffic environments.

2. Introduction

Traffic congestion has become one of the most significant challenges facing modern cities as a result of rapid urbanization and the growing number of vehicles on the road. It leads to longer travel times, increased fuel consumption, higher transportation costs, and increased air pollution, while also reducing the overall efficiency of urban transportation systems. To address these challenges, governments and organizations are increasingly adopting intelligent technologies that enhance traffic management and support the development of smart cities.

Artificial Intelligence (AI), together with technologies such as IoT, Graph Neural Networks (GNN), Long Short-Term Memory (LSTM), reinforcement learning, and Explainable Artificial Intelligence (XAI), has significantly improved the ability of intelligent transportation systems to predict congestion, optimize traffic signals, and recommend efficient travel routes. Intelligent traffic management plays an important role in improving transportation efficiency, supporting logistics and supply chain operations, reducing costs and enhancing customer satisfaction through shorter and more reliable travel times. These improvements contribute to increased productivity and support the development of sustainable smart city infrastructure.

The purpose of this literature review is to examine recent research on AI-based intelligent traffic management systems. It compares the methods, findings, and limitations of selected studies to identify current research trends, highlight common challenges, and discuss opportunities for future research in intelligent traffic forecasting, congestion prediction, and smart transportation systems.

3. Methodology

The literature search for this review was conducted using Google Scholar to identify recent peer-reviewed studies related to artificial intelligence in intelligent traffic management. Google Scholar was chosen since it provides access to reliable research published by well-known publishers such as Springer, IEEE, Elsevier, etc.

The search used keywords related to intelligent traffic management, traffic prediction, machine learning, deep learning, IoT, GNN, LSTM, reinforcement learning, XAI, and smart transportation. These keywords were combined using different search queries to identify studies focusing on traffic forecasting, congestion prediction, route optimization, and AI-based traffic management systems. The search results were reviewed, and the most relevant studies were selected based on the predefined inclusion criteria.

The selection of papers was based on several conditions. Only studies published between 2021 and 2025 were considered to ensure that the review reflects recent developments in the field. The selected studies were directly related to the keywords and had to be available through Google Scholar with accessible publication information.

The final selection consisted of ten research papers that addressed different aspects of intelligent traffic management. The studies included quantitative research, systematic reviews, and experimental studies. Together, they covered a range of artificial intelligence techniques.

Figures 1-4 present examples of the Google Scholar search results and the selected papers used in this review.

Figure 1. Google Scholar search result for "Smart transportation with machine learning and Internet of Things IoT".

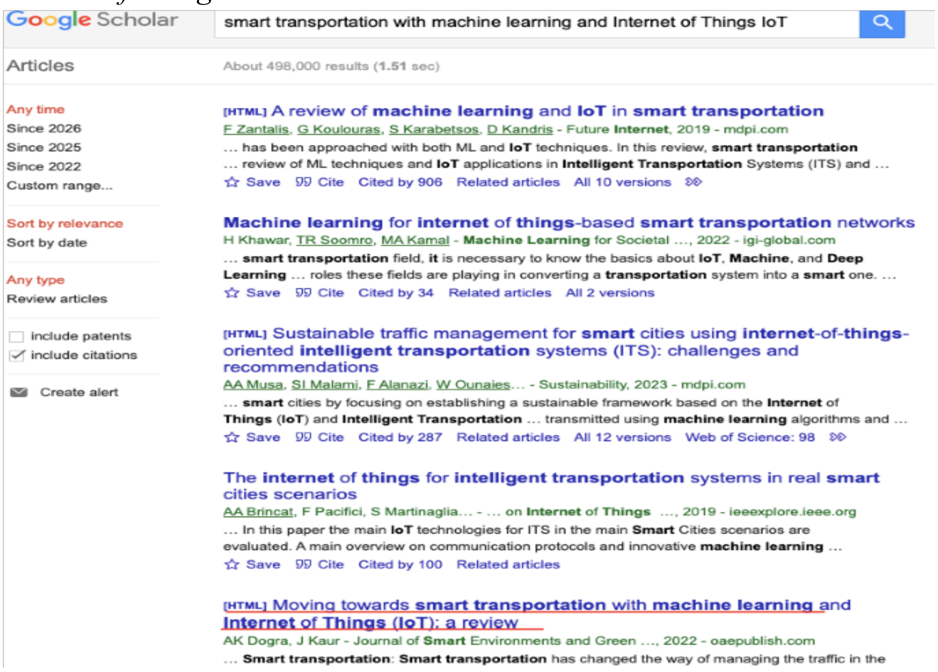


Figure 2. Google Scholar search result for "Traffic Solutions Reinforcement Learning and Explainable AI".

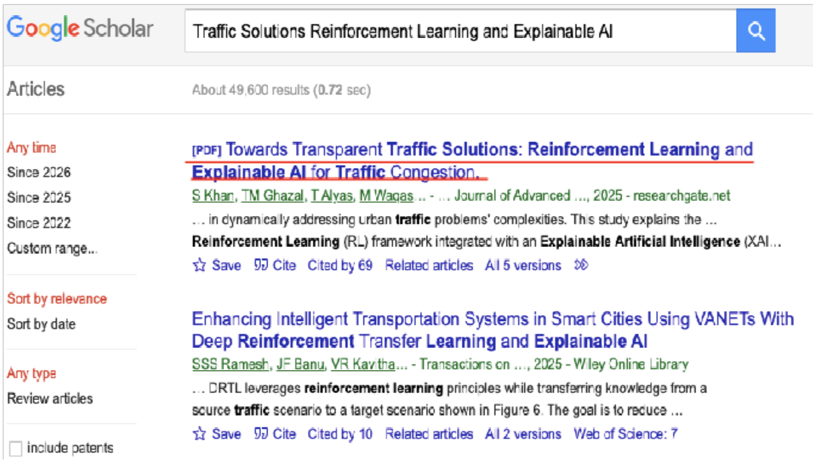


Figure 3. Google Scholar search result for "Traffic flow prediction for smart traffic".

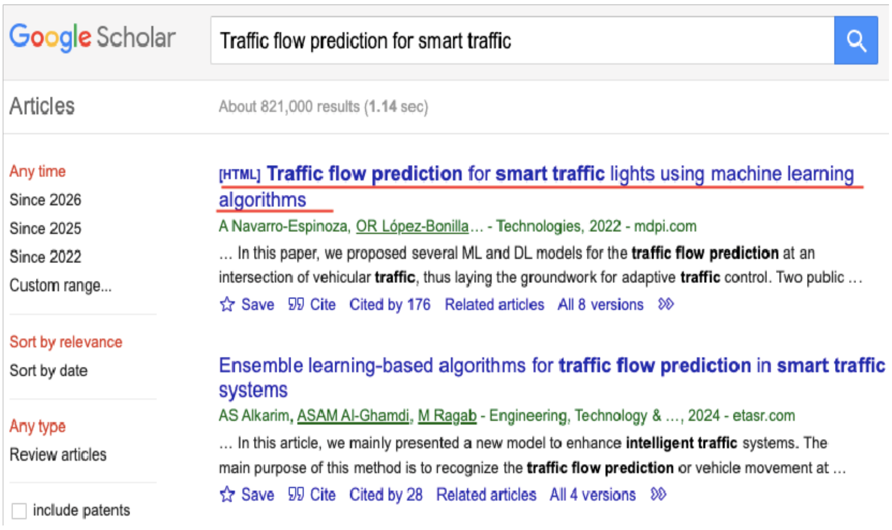
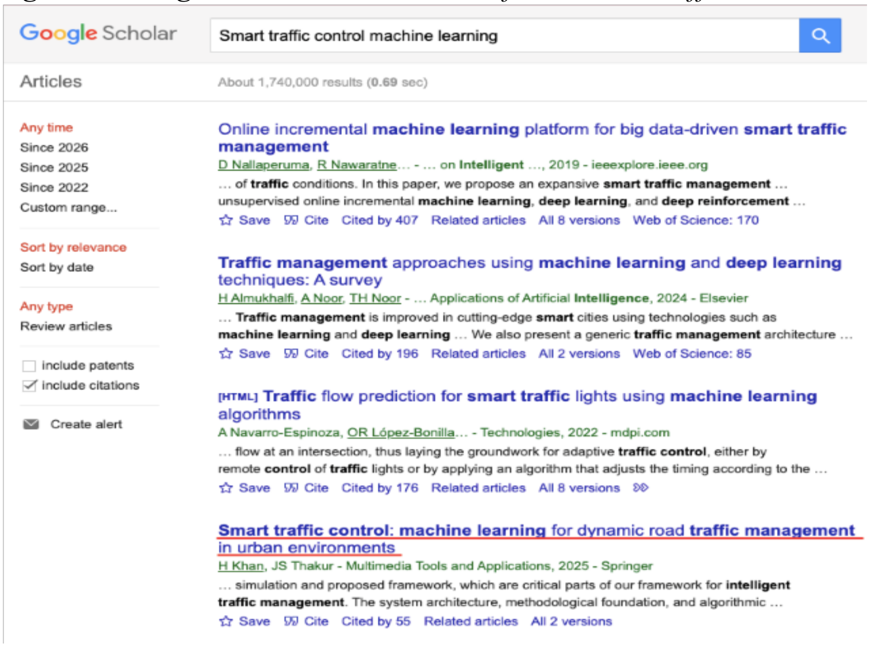


Figure 4. Google Scholar search result for "Smart traffic control machine learning".



4. Results and Discussion

4.1 Overview of the Selected Studies

This section presents a comparative analysis of the selected studies. The papers were reviewed to identify the main research objectives, methodologies, findings, and limitations. Table 1 provides a summary of the studies, followed by a discussion of the common trends, differences, and research gaps identified across the literature.

Table1. Summary of the Selected Studies

Ref.	Year	Objective	Methodology	Results	Limitations
[1]	**2025**	**Improve dynamic traffic management**	**Machine Learning**	**Improved traffic flow and congestion prediction**	**Depends on high quality traffic data**
[2]	**2023**	**Intelligent traffic management using IoT and ML**	**IoT + ML**	**Better congestion monitoring**	**Requires IoT infrastructure**
[3]	**2023**	**Secure traffic management and accident detection**	**IoT + ML**	**Improved accident detection**	**High implementation cost**
[4]	**2022**	**Review ML and IoT in transportation**	**Literature review**	**Summarized current AI applications**	**General review only**
[5]	**2023**	**Review intelligent transportation systems**	**Literature review**	**Identified ITS trends**	**Mostly conceptual**
[6]	**2025**	**Prioritize emergency vehicles**	**IoT + UWB technology**	**Reduced emergency response time**	**Simulation based validation**
[7]	**2023**	**Explainable GNN traffic forecasting**	**GNN + XAI**	**Improved interpretability**	**Computational complexity**
[8]	**2025**	**Transparent traffic congestion prediction**	**Reinforcement learning + XAI**	**Better adaptive traffic control**	**Long training time**
[9]	**2023**	**Accident prediction**	**ML comparison**	**Improved prediction performance**	**Regional dataset**
[10]	**2022**	**Smart traffic light prediction**	**Machine Learning**	**Better traffic flow prediction**	**Limited datasets**

4.2 Artificial Intelligence in Intelligent Traffic Management

The reviewed studies demonstrate the growing role of AI in improving intelligent traffic management. Traditional traffic management systems mainly respond to congestion after it occurs, whereas AI-based approaches enable traffic conditions to be predicted in advance, allowing transportation authorities to make proactive decisions. Most of the studies applied AI to improve traffic prediction, congestion management, accident detection, or route optimization.

Machine learning was the most frequently used approach because it can efficiently analyze large volumes of traffic data and identify patterns that are difficult to detect using conventional methods. Review studies such as [4] also emphasize that combining machine learning with IoT technologies has become one of the key drivers of modern intelligent transportation systems. Studies by Khan and Thakur [1], Atassi and Sharma [2], and Navarro-Espinoza et al [10] reported that AI models improved prediction accuracy and supported more efficient traffic management compared to traditional methods. Although the reviewed studies used different datasets and algorithms, they all showed that AI contributes to more intelligent and data-driven transportation systems.

4.3 Machine Learning and Deep Learning Techniques

The papers applied a variety of machine learning and deep learning techniques depending on the research objective. Traditional machine learning algorithms were commonly used for congestion prediction and traffic classification because they are relatively simple to train and provide reliable performance on structured traffic datasets. In contrast, deep learning techniques were mainly applied to traffic forecasting because they are better suited for analyzing sequential traffic data.

Some of the studies also introduced more advanced approaches. Garcia-Siguenza et al [7] combined GNN with XAI to improve both prediction performance and model interpretability. Similarly, Khan and Patel [8] integrated reinforcement learning with explainability techniques to develop adaptive traffic management systems capable of responding to changing traffic conditions. These studies demonstrate a shift from developing highly accurate prediction models toward creating systems that are also transparent and easier to understand. In addition, Berhanu et al. [9] showed that machine learning techniques can be effectively applied to accident prediction, demonstrating that AI supports not only traffic forecasting and congestion management but also road safety by identifying accident risks before they occur.

Although advanced models generally achieved higher prediction accuracy, they also required greater computational resources and larger datasets than traditional machine learning algorithms.

4.4 IoT- Based Smart Traffic Management

Internet of Things technology was another common component of the studies. IoT devices such as sensors, cameras, roadside units, and communication systems enable continuous collection of real-time traffic information. The literature survey presented in [5] highlights that roadside infrastructure plays a critical role in enabling communication between vehicles and intelligent traffic management systems.

Atassi and Sharma [2], Balasubramanian and Balaji [3], and Fida et al. [6] demonstrated how IoT technologies improve traffic monitoring, congestion management, accident detection, and emergency vehicle prioritization. These studies showed that combining IoT with AI enables faster decision-making and improves overall transportation efficiency. However, they also highlighted practical challenges, including high implementation costs, infrastructure requirements, communication reliability, and ongoing maintenance.

4.5 Current Challenges and Research Gaps

Despite the promising results reported across the papers, several common challenges remain. Most AI models rely on large volumes of accurate and continuously updated traffic data. Missing sensor readings, noisy datasets, and incomplete information can significantly reduce prediction accuracy. In addition, advanced models such as GNN and reinforcement learning require substantial computational resources, making them more difficult to deploy in largescale transportation systems.

Another common limitation is that many studies evaluated their models using simulation or publicly available datasets rather than real-world traffic environments. This limits the ability to assess how these systems would perform under changing road conditions, unexpected events, or infrastructure constraints. Furthermore, only a few studies focused on model explainability, even though transparent AI systems are becoming increasingly important for supporting decision-making in smart cities.

Future research should focus on improving the scalability of AI-based traffic management systems, increasing the availability of high-quality real-world datasets, and developing more interpretable models that can be deployed efficiently in practical transportation environments.

5. Conclusion

This literature review examined recent research on the application of artificial intelligence in intelligent traffic management, with a particular focus on machine learning, deep learning, internet of things, graph neural networks, reinforcement learning, and explainable AI. The reviewed studies showed that AI-based approaches have significantly improved traffic forecasting, congestion prediction, accident detection, and traffic signal optimization compared with traditional traffic management methods.

Although different techniques were used across the selected studies, they all demonstrated the potential of AI to support more efficient, accurate, and data-driven transportation systems. The review also highlighted the important role of IoT technologies in providing real-time traffic data, enabling faster and more informed decision-making.

Despite these advancements, several challenges remain. Many AI models depend on high-quality real-time data, advanced computing resources, and reliable communication infrastructure. In addition, several studies were evaluated using simulation environments or public datasets, making further real-world validation necessary.

Future research should focus on developing more scalable and interpretable AI models while improving data quality and increasing the real-world implementation of intelligent traffic management systems. Addressing these challenges will contribute to more efficient, sustainable, and reliable transportation systems that support the continued development of smart cities.

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